Fiscal Year:	FY 2015	**Task Last Updated:**	FY 09/22/2015
PI Name:	Mollicone, Daniel Ph.D.		
Project Title:	STARwatch to Deliver Objective Sleep Measures for Spaceflight Operations		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	06/16/2015	**End Date:**	06/15/2017
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/18/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX15CJ05C		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project will achieve a wrist-worn actigraphy device called STARwatch, designed specifically for space exploration environments. It will provide a minimally obtrusive, objective measure that evaluates astronaut sleep-wake activity and light exposure. This project will leverage our second-generation actigraphy device that has already been validated in controlled laboratory experiments against gold-standard polysomnography. The compact wrist-worn device includes sensors to collect sleep metrics and will also serve as a wireless hub to collect real-time physiological data from other body-worn sensors (e.g., heart rate, EEG). It will use standardized wireless communication protocols (e.g., Bluetooth) to automatically uplink data to the International Space Station (ISS) network (no astronaut time required). Data will automatically be integrated into medical operations support systems adhering to NASA data requirements (e.g., HL7), providing immediate feedback to astronauts and flight surgeons to aid in decision-making relative to astronaut medical, behavioral health and performance issues. During Phase II, we will conduct user testing and validation in a space flight analog environment, complete product refinements, and certify STARwatch for spaceflight. (Phase II TRL of 7-8). Currently in spaceflight, crewmembers report their sleep duration as requested by their crew surgeon. This approach has several limitations, including the burden it places on the crew and the tendency for subjective over-reporting of sleep (Lauderdale et al., 2008; Van Den Berg et al., 2008; Silva et al., 2007). Given evidence that demonstrates the relationship between sleep and circadian phase and performance, sleep-activity data should be collected as unobtrusively possible during long duration spaceflight. Wrist-worn actigraphy has been implemented as a successful, validated research tool in spaceflight but lacks features to render it a useful tool operationally, such as real-time feedback and minimal crew time requirements. Hence, there is a need for a minimally obtrusive or unobtrusive measure that evaluates sleep-wake activity plus light exposure; is acceptable for continuous wear; minimizes crew time by allowing for automatic downloads; provides immediate feedback to the user; incorporates the constraints of spaceflight hardware, such as extended battery life; and potentially incorporates other features, including other physiological sensors. SBIR Phase II: The purpose of this task is to test and validate refined technologies so that they serve as an objective measure of sleep-wake activity (and lighting) while fitting into the spaceflight mission environment. The Technical Objectives of the Phase II project are: 1.) Conduct user testing to achieve initial product validation in a space analog environment and guide Phase II product refinement. 2.) Develop an updated version of STARwatch tailored to meet the exact needs of space exploration. 3.) Develop state-of-the-art sensing and advanced analytics features to support space exploration missions. 4.) Conduct user testing with the updated version of STARwatch to validate updated features and prepare for transition to space flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There is an articulated market need for tools that track sleep and performance, particularly in areas where human performance has precise operational constraints and important safety implications, such as commercial aviation, emergency health care, shift work, etc. Sleep monitoring technologies described in this proposal can help meet this need by providing professionals working in safety-sensitive occupations with immediate feedback about their sleep and performance and assist with selecting fatigue countermeasures.
Task Progress:	New project for FY2015. Reporting not required for this SBIR Phase 2 project. [Ed. note: added to Task Book when received information on the task in September 2015]
Bibliography Type:	Description: (Last Updated: 02/23/2015)